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Chapter 1

Topology

1.1 Topology

A set $N \subset S$ is called a neighbourhood of $x \in S$ if it contains a ball $B_r(x)$. A set is open if it is a neighbourhood of all its points. A point $x \in X$ is an isolated point if it has a neighbourhood whose intersection with X reduces to x. An accumulation point is a point that is not isolated.

Proposition 1.1. *A non-empty open set in plane is connected if and only if any two points can be joined by a polygon which lies in the set.*

Definition: A non-empty connected open set is called a region. A component or a maximal connected set of a set S is a connected subset which is not contained in any larger connected subset.

Proposition 1.2. *Every set has a unique decomposition into component.*

Proposition 1.3. *In $\mathbb{R}^n$ the components of any open set are open.*

Definition: A set A is dense in X if $\operatorname{cl} E = X$. A metric space is separable if there exists a countable subset union of disjoint regions. A topological space is locally connected if for each neighbourhood of its points there is a connected open subset for that neighbourhood. A metric space that open balls $B_r(x)$ are connected is locally connected.

Proposition 1.4. *In a locally connected separable space every open set is union of disjoint regions.*

Definition: A set S is totally bounded if for every $\epsilon > 0$, S can be covered by finitely many balls of radius ϵ.

1.2 Compact sets

Definition (Point of accumulation): point v is a **point of accumulation** for the sequence $\{z_n\}$ if for given $\epsilon > 0$ there exists infinitely many n such that

$$|z_n - v| < \epsilon$$

Similarly, a point of accumulation of an infinite set S is a point v that for each open set U containing v there are infinitely many elements of S.

Theorem 1.5 (Weierstrass-Bolzano theorem). *If S is an infinite bounded set of real numbers, then S has a point of accumulation.*

Definition: A **compact** set S is a set that every sequence of its elements has a point of accumulation in S. The following defintions are equivalent

1. Every infinite subset of S has a point of accumulation in S.
2. Every sequence of elements of S has a convergent subsequent whose limit is in S.

Theorem 1.6. *A complex set is compact if and only if it is closed and bounded.*

Theorem 1.7. *Let $S_1 \supset S_2 \supset \dots$ be a sequence of non-empty closed subsets of a compact set S. Then, the interestion of all S_n is not empty.*

Theorem 1.8. *Let S be a compact set and f be continuous function on S. Then the image of f is compact.*

Theorem 1.9. *Let S be a compact set and f be continuous function on S. Then f is uniformly continuous.*

Definition: Let A, B be two sets of complex numbers. The **distance** between them is

$$d(A, B) = \min_{\substack{\alpha \in A \\ \beta \in B}} |\alpha - \beta|$$

Theorem 1.10. *Let S be a closed set and let v be a complex number. There exists a point $w \in S$ such that*

$$d(S, \{v\}) = |w - v|$$

Theorem 1.11. *Let K be a compact set and let be S a closed set. Then, there are elements $\alpha \in K$ and $\beta \in S$ such that*

$$d(K, S) = |\alpha - \beta|$$

Theorem 1.12. *A set is compact if and only if it is complete and totally bounded.*

Corollary 1.13. *Let K be compact. Let r be a real number greater than zero. There exists a finite number of discs of radius r whose union contains K.*

We say that a family of open set $\{U_i\}$ covers a set S when for every $z \in S$, $z \in U_i$ for some i as well. A subcovering of S is a covering of S with a subfamily of $\{U_i\}$. If that subfamily is finite we say that it is a finite subcovering of S. S is **covering compact** if every open convering can reduce to a finite subcovering.

Theorem 1.14. *Let S be a set then S is sequentially comapct if and only if covering compact.*

1.3 Connectedness

Proposition 1.15. *A path connected set is connected but the converse is true when the set is open.*

Chapter 2

Complex Numbers

2.1 Algebra of Complex Numbers

We define complex numbers to be all the pairs of real numbers (x, y) with following addition and multiplication:

$$(x_1, y_1) +' (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$(x_1, y_1) \cdot' (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1)$$

where $+$ and $\cdot$ are real addition and multiplication, respectively. We denote the set of complex numbers with $\mathbb{C}$. It is easy to check that addition and multiplication defined above have the following:

1. Addition and multiplication are commutitive:

$$z + w = w + z \quad z, w \in \mathbb{C}$$

$$z \cdot w = w \cdot z \quad z, w \in \mathbb{C}$$

2. Addition and multiplication are associative:

$$(z + w) + u = z + (w + u) \quad z, w, u \in \mathbb{C}$$

$$(z \cdot w) \cdot u = z \cdot (w \cdot u) \quad z, w, u \in \mathbb{C}$$

3. Addition and multiplication are distributive:

$$(z + w) \cdot u = z \cdot u + w \cdot u \quad z, w, u \in \mathbb{C}$$

4. Addition and multiplication have unique identity elements $0 = (0, 0)$ and $1 = (1, 0)$, respectively.

5. Every complex number z has a unique addition inverse. Denoted by $-z$.

6. Every non-zero complex number z has a unique multiplication inverse. Denoted by z^{-1} or $\dfrac{1}{z}$.

Which means $\mathbb{C}$ is a field.

We can represent in many other forms. Two of the most commonly for $z = (x, y)$ used are :

$$z = x + iy \quad \text{where } i = (0, 1)$$
$$z = re^{i\theta} \quad r \geq 0\,, \theta \in \mathbb{R}$$
$$z = \begin{bmatrix} x & -y \\ y & x \end{bmatrix}$$

It is easy to see that $i^2 = -1$. We also define the functions $\Re(z) = x$ and $\Im(z) = y$. In the second representation r is the distance from origin and θ is the angle between the positive real axis and the ray passing through z. One can also view complex number as an extension of real numbers in which every polynomial has a root.

Theorem 2.1. *If K is field such that every odd degree polynomial has at a least a root and for all $a \in K$, either $x^2 = a$ or $x^2 = -a$ has a root then it is sufficient to add the root of $x^2 = -1$ to K so that every polynomial has a root.*

For every complex number $z = x+iy$ there exists the mapping $\bar{z} : \mathbb{C} \to \mathbb{C}$ where $\bar{z} = x-iy$ and is called the *conjugate* of z.

Proposition 2.2. *The following properties are satisfied.*

1. $\overline{z + w} = \bar{z} + \bar{w}$.
2. $\overline{zw} = \bar{z}\bar{w}$.
3. $z = \bar{\bar{z}}$.
4. $z = \bar{z}$ *if and only if* $z \in \mathbb{R}$.
5. *The following relations hold:*
$$\Re(z) = \frac{z + \bar{z}}{2} \quad , \quad \Im(z) = \frac{z - \bar{z}}{2i}$$

Another mapping is the *norm* or *modulus* function, $|z| : \mathbb{C} \to \mathbb{R}$ where $|z| = \sqrt{x^2 + y^2}$. Geometrically speaking the norm of z gives the distance of z form origin.

Proposition 2.3. *The following properties are satisfied.*

1. $|z| \geq 0 \quad \forall z \in \mathbb{C}$ *and especially* $|z| = 0$ *if and only if* $z = 0$
2. $|zw| = |z||w|$.
3. $|z|^2 = z\bar{z}$.
4. $|z| = |-z| = |\bar{z}|$
5. *The following inequalities hold:*
$$-|z| \leq \Re(z)\,,\ \Im(z) \leq |z|$$
6. *Triangle inequality:*
$$|z + w| \leq |z| + |w|$$
$$||z| - |w|| \leq |z - w|$$

The point at infinity, ∞, is informally a point that is unboundedly far from the origin. The extended complex plane is defined as $\mathbb{C} \cup \{\infty\}$. Every line passes through ∞ but no half plane contains it.

2.2 Riemann sphere

Riemann sphere is a unit sphere centered at the origin complex plane. We can map every point (except $(0,0,1)$) on the sphere to a point in complex plane by the following bijective transformation.

$$\phi(z) = (x_1, x_2, x_3) = \left(\frac{z+\bar{z}}{1+|z|^2}, \frac{z-\bar{z}}{i\big(1+|z|^2\big)}, \frac{|z|^2-1}{|z|^2+1} \right)$$

Geometrically, this transformation maps every point z to the interestion of the line connecting it to the $(0,0,1)$ and the sphere. Furthermore, we can define $\phi\infty(=)(0,0,1)$.

2.3 Limits and continuity

Let $f : U \to \mathbb{C}$, α be an adherent point of U, and w be a complex number. Then

$$w = \lim_{z\to\alpha} f(z)$$

when

$$\forall\epsilon > 0,\ \exists\delta > 0 \text{ s.t. } z \in U, |z-\alpha| < \delta \implies |f(z) - w| < \epsilon$$

By defining the $\epsilon-$neighbourhood of ∞ to the

$$\left\{ z \,\middle|\, \frac{1}{z} < \epsilon \right\}$$

we can extend the defintion of limit to when α or w are ∞. We say that f is continuous at α if

$$\lim_{z\to\alpha} f(z) = f(\alpha)$$

Definition: A function f is said to be uniformly continuous if

$$\forall\epsilon > 0,\ \exists\delta > 0 \text{ s.t. } z, w \in S, |z-w| < \delta \implies |f(z) - f(w)|$$

2.4 Sequences and convergence

A Sequence $\{z_n\}_{n\in\mathbb{N}}$ is said to be convergent to z if:

$$\forall\epsilon > 0,\ \exists N \text{ s.t. } n \geq N \implies |z - z_n| < \epsilon$$

and it is denoted as

$$z = \lim_{n\to\infty} z_n$$

A sequence $\{z_n\}$ is a **Cauchy sequence** if

$$\forall\epsilon,\ \exists N \text{ s.t. } n, m \geq N \implies |z_m - z_n| < \epsilon$$

If we write $z_n = x_n + iy_n$, since $|z_n - z_m| = \sqrt{(x_n - x_m)^2 + (y_n - y_m)^2}$ and $|x_n - x_m| \leq |z_n - z_m|$, $|y_n - y_m| \leq |z_n - z_m|$, we can conclude that $\{z_n\}$ is Cauchy if and only if $\{x_n\}$ and $\{y_n\}$ are Cauchy sequences. Thus, since for real sequences a Cauchy sequence is convergent then, a complex Cauchy sequence is convergent as well.

Due to similarities between the definition given above and their real counterpart, most results dealing with limits can be easily extended to complex numbers.

A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ is convergent.

Proposition 2.4. *If the series $\sum a_n$ is absolutely convergent then it can be summed in any order.*

Proposition 2.5. *If double sum $\sum_m \sum_n a_{mn}$ is absolutely convergent then the summation order can be interchanged.*

$$\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$$

The resulting series obtained is absolutely convergent and converges to the same value.

Definition: Let $\{a_n\}$ and $\{b_n\}$ be two sequences of positive real numbers. Then

$$a_n \equiv b_n \qquad \text{for} \quad n \to \infty$$

if for each n, there exists a $u_n \in \mathbb{R}^+$ such that $\lim_{n\to\infty} u_n^{\frac{1}{n}} = 1$ and $a_n = b_n u_n$.

2.5 Function spaces and power series

A family of functions $\{f_n(x) : X \to Y\}$ converges uniformly to $f : X \to Y$, denoted $f_n \rightrightarrows f$

$$\forall \epsilon,\ \exists N \text{ s.t. } n \geq N \implies d_Y(f_n(x) - f(x)) < \epsilon,\ \forall x$$

if Y is a complete metric space then Cauchy convergence becomes equivalent to uniform convergence

$$\forall \epsilon,\ \exists N \text{ s.t. } n, m \geq N \implies d_Y(f_n(x) - f_m(x)) < \epsilon,\ \forall x$$

$\sum f_n(x)$ converges uniformly if the partial sums $s_m = \sum^m f_n$ converges uniformly.

Theorem 2.6 (Weierstrass M-test). *Suppose that $\{f_n(x)\}$ is a sequence of real/complex-valued functions defined on a set X, and there is a sequence of non-negative numbers M_n satisfying the conditions*

- *$|f_n(x)| \leq M_n$ for all $n \geq 1$ and all $x \in X$*
- *$\sum_{n=1}^{\infty} M_n$ converges*

then the series $\sum_{n=1}^{\infty} f_n(x)$ converges absolutely and uniformly on X.

Theorem 2.7. *Let $\{f_n(x)\}$ be a sequence of continuous functions that $f_n \rightrightarrows f$. Then f is continuous.*

Chapter 3

Analytic Functions

3.1 Differentiability

Let $f : U \overset{open}{\subset} \mathbb{C} \to \mathbb{C}$ and z be a point in U. f is said to be complex differentiable at z if the following limit exists.

$$f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h}$$

Equivalently, there exists a $c \in \mathbb{C}$ such that

$$f(z_0 + h) - f(z_0) - hc = o(h) \tag{3.1}$$

Differentiation rules are similar to real differentiation and are proved the same way.

f is **holomorphic** if it is differentiable at every point of U. Taking the limit along the real and imaginary axes and equating them gives the **Cauchy-Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Assuming that u and v are twice differentiable, we can see that

$$\begin{aligned} \Delta u &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \\ &= \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x} = 0 \end{aligned}$$

Which means that u, v satisfy the *Laplace's equation* and thus are harmonic functions.

Theorem 3.1. *If u, v and have continuous first order partial derivatives and satisfy the Cauchy-Riemann equations then $f(z) = u(x, y) + v(x, y)$ is analytic with continuous derivative.*

3.2 Linear transformation

A linear fractional transformation, also called *Möbius transformation*, is a function in form of

$$w = S(z) = \frac{az + b}{cz + d}$$

with $ad - bc \neq 0$. If $c \neq 0$ then $S(\infty) = \frac{a}{c}$ and $S\left(-\frac{d}{c}\right) = \infty$. The inverse of a linear fractional transformation is linear fractional itself and is equal to

$$S^{-1}(w) = \frac{dw - b}{-cw + a}$$

If $ad - bc = 1$ we say that S is normalized and since multiplying a, b, c, d by a non-zero factor does not change the function we can assume that all Möbius transformations are normalized.

Theorem 3.2. *The set of all Möbius transformations $\mathcal{M}$ is a group under composition.*

$$\mathcal{M} = \left\{ \frac{az + b}{cz + d} \,\middle|\, ad - bc = 1, a, b, c, d \in \mathbb{C} \right\}$$

Proposition 3.3. *Give 3 different points z_1, z_2, z_3 there exists a linear transformation that takes to $1, 0, \infty$ in that order. If non of the point is ∞ then*

$$S(z) = \frac{z - z_2}{z - z_3} \frac{z_1 - z_3}{z_1 - z_3}$$

if one of them is ∞ then

$$S(z) = \frac{z - z_2}{z - z_3}, \quad S(z) = \frac{z_1 - z_3}{z - z_3}, \quad S(z) = \frac{z - z_2}{z_1 - z_2}$$

for $z_1 = \infty, z_2 = \infty, z_3 = \infty$ respectively.

Proof. It is clear that S as defined above is such a transformation. To prove S is unique, suppose there is another transformation T that satisfies the condition. Suppose that neither of the points is ∞ and

$$T(z) = \frac{az + b}{cz + d}$$

for some $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$. Consider TS^{-1}

$$\begin{aligned} TS^{-1}(z) &= \frac{aS^{-1}(z) + b}{cS^{-1}(z) + d} \\ &= \frac{a(z - z_2)(z_1 - z_3) + b(z - z_3)(z_1 - z_2)}{c(z - z_2)(z_1 - z_3) + d(z - z_3)(z_1 - z_2)} \\ &= \frac{(az_1 + bz_1 - az_3 - bz_2)z + az_2z_3 - az_2z_1 + bz_3z_2 - bz_3z_1}{(cz_1 + dz_1 - cz_3 - dz_2)z + cz_2z_3 - cz_2z_1 + dz_3z_2 - dz_3z_1} \end{aligned}$$

We know that

$$TS^{-1}(1) = 1, \quad TS^{-1}(1) = 1, \quad TS^{-1}(\infty) = \infty,$$

which implies that

$$\begin{cases} az_2z_3 - az_2z_1 + bz_3z_2 - bz_3z_1 = 0 \\ cz_1 + dz_1 - cz_3 - dz_2 = 0 \\ az_1 + bz_1 - az_3 - bz_2 = cz_2z_3 - cz_2z_1 + dz_3z_2 - dz_3z_1 \end{cases}$$

hence

$$TS^{-1}(z) = \frac{(az_1 + bz_1 - az_3 - bz_2)z}{az_1 + bz_1 - az_3 - bz_2} = z$$

and thus TS^{-1} is the identity and we must have $T = S$. ■

In general we have the following

Proposition 3.4. *For any two sets of distinct complex numbers z_1, z_2, z_3 and w_1, w_2, w_3 there exists a linear fractional transformation that takes z_i to w_i for $i = 1, 2, 3$.*

Proof. We first show that following lemma

Lemma 3.5. *Any linear fractional transformation has at most two fixed points unless it is the identity.*

Proof. It is clear that

$$\frac{az+b}{cz+d} = z \implies cz^2 + (d-a)z + b = 0$$

has at most two solutions unless $c = 0$ and $d - a = 0$. □

Then consider the following linear fractional transformation (interpreted appropriately if any of them is ∞)

$$S = w_1 \frac{(z-z_2)(z-z_3)}{(z_1-z_2)(z_1-z_3)} + w_2 \frac{(z-z_1)(z-z_3)}{(z_2-z_1)(z_2-z_3)} + w_3 \frac{(z-z_1)(z-z_2)}{(z_3-z_1)(z_3-z_1)}$$

Clearly S the required property. Suppose there exists another linear fractional transformation T with such property. We have

$$TS^{-1}w_i = w_i, \quad i = 1, 2, 3$$

which means TS^{-1} is the identity and we must have $T = S$. ■

Definition (Cross ratio): The cross ratio of $(z_1, z_2; z_3, z_4)$ is the image of z_1 under the transformation S that takes z_1, z_2, z_3 to $1, 0, \infty$. That is

$$(z_1, z_2; z_3, z_3) = \frac{z_1 - z_3}{z_1 - z_4} \frac{z_2 - z_4}{z_2 - z_3}$$

Proposition 3.6. *Let $z_1, z_2, z_3, z_4 \in \mathbb{C}$ and T is a linear fractional transformation. Then $(Tz_1, Tz_2; Tz_3, Tz_4) = (z_1, z_2; z_3, z_3)$*

Proof. Suppose $S = (z, z_2; z_3, z_4)$ and $S' = (z, Tz_2; Tz_3, Tz_4)$. Note that

$$ST(z_2) = 1, \quad ST(z_3) = 0, \quad ST(z_4) = \infty$$

therefore $S = S'T$ and for all $z_1 \in \mathbb{C}$, $Sz = S'Tz$. ■

Proposition 3.7. *Let $r, c \in \mathbb{R}$ and $k \in \mathbb{C}$, then the equation*

$$r|z|^2 + k\bar{z} + \bar{k}z + c = 0$$

represents a line if $r = 0$, represents a circle if $r \neq 0$ and $|k|^2 \geq rc$.

The locus of all the points of $r|z|^2 + k\bar{z} + \bar{k}z + c = 0$, if non-empty, is called a circline.

Proposition 3.8. *circlines correspond to circles on Riemann sphere.*

Proposition 3.9. *A linear transformation carries circlines to circlines.*

Proposition 3.10. *The cross ratio is real if and only if the four points lie in a circle or a straight line.*

3.3 Polynomials and rational functions

Theorem 3.11. *If all zero's of a polynomial P lie in a half plane (convex polygon), the all zero's of the derivative P' lie in the same half plane (convex polygon).*

A **rational function** $R(z)$ is the quotient of two polynomials

$$R(z) = \frac{P(z)}{Q(z)} = \frac{a_m x^m + \cdots + a_0}{b_n x^n + \cdots + b_0}$$

We can assume that P and Q have no factor in common. In that case, the zeros of P are the zeros of R and the zeros of Q are called the *poles* of R. We can further define $R(\infty)$ to be $S_R(0)$ where $S_R(z) = R\big(\frac{1}{z}\big)$.

$$S_R(z) = z^{n-m} \frac{a_m + \cdots + a_0 x^m}{b_n + \cdots + b_0 x^n}$$

Then if $n > m$ we say that R has roots at ∞ with multiplicity of $n - m$, if $n < m$ then R has poles at ∞ with multiplicity of $m - n$. If $n = m$ then $R(\infty) = \frac{a_m}{b_n}$. It is now easy to see that the number of roots and poles of R - including ∞ - is the greater of m, n and it is called the *degree* of R.

We can write R in the following form

$$R(z) = G(z) + H(z)$$

where G is a polynomial with no constant term, by carrying out the division. Now let $\beta_1, \ldots, \beta_k$ be distinct poles of R. Then $S_j(\xi) = R\Big(\beta_j + \frac{1}{\xi}\Big)$ is a rational function of ξ and it can be written as

$$S_j(\xi) = G_j(\xi) + H_j(\xi) \implies R(z) = G_j\left(\frac{1}{z - \beta_j}\right) + H_j\left(\frac{1}{z - \beta_j}\right)$$

note that $G_j\Big(\frac{1}{z-\beta_j}\Big)$ has pole(s) only at β_j. Now consider

$$S(z) = R(z) - G(z) - \sum_{j=1}^{k} G_j\left(\frac{1}{z - \beta_j}\right)$$

It can have poles only at $\beta_1, \ldots, \beta_k$ and ∞. R and G_j have poles at $z = \beta_j$, however their difference if finite. R and G have poles at ∞ but their difference is finite as well. Hence S has no poles and thus must be a constant. Incorporating the constant into G allows us to write R as

$$R(z) = G(z) + \sum_{j=1}^{k} G_j\left(\frac{1}{z - \beta_j}\right)$$

which is its *partial fraction decomposition.*

3.4 Conformal maps

A **conformal map** is a transformation that conserves angle and direction. For example every linear $\mathbb{R}^2$ transformation of the form

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \sqrt{a^2+b^2} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

is a conformal mapping since it is a scaling and a rotation, both of which conserve the angle and direction. Let U be an open set in $\mathbb{C}$ and $\gamma, \eta : [a, b] \to U$ be two complex valued curves. Suppose that $z_1 = \gamma(t_1) = \eta(t_2)$ then the angle between γ and η at z_1 is defined as the angle between $\gamma'(t_1)$ and $\eta'(t_2)$, provided that they are nonvanishing. Let $f : U \to \mathbb{C}$ be a holomorphic function then by chain rule

$$\frac{\mathrm{d}f(\gamma(t))}{\mathrm{d}t} = f'(\gamma(t))\gamma'(t)$$

Viewing $w = f'(\gamma(t))$ as a $\mathbb{R}^2$ linear transformation, we can deduce that f is conformal.

$$\frac{\mathrm{d}f(\gamma(t_1))}{\mathrm{d}t} = f'(z_1)\gamma'(t_1)$$
$$\frac{\mathrm{d}f(\eta(t_2))}{\mathrm{d}t} = f'(z_1)\eta'(t_2)$$

Thus given that $f'(z_1) \neq 0$ the angle between γ and η at z_1 is the same as $f \circ \gamma$ and $f \circ \eta$.

Chapter 4

Formal Power Series

4.1 Algbraic definitions

Monoid

A set S equipped with a binary operation $\cdot$ is a **monoid** if it satisfies the following properties:

- For all $a, b, c \in S$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- There exists $e \in S$ such that for each $a \in S$, $a \cdot e = e \cdot a = a$.

Group

A set S equipped with a binary operation $\cdots$ is a **group** if it satisfies the following properties:

- For all $a, b, c \in S$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- There exists $e \in S$ such that for each $a \in S$, $a \cdot e = e \cdot a = a$.
- For each element $a \in S$, there exists $b \in S$ such that $a \cdot b = b \cdot a = e$.

If $\cdot$ is commutative then the group is an *Abelian group*.

Ring

A set S equipped with binary operations $+$ and $\cdots$ is a **ring** if it satisfies the folloing properties:

1. S is an Abelian group under $+$.
2. S is a monoid under $\cdot$.
3. $\cdot$ distributes over $+$.

If $\cdot$ is commutative then the ring is called a *commutative ring*.

Integral domain

It is commutative ring with property that

$$ab = 0 \implies a = 0 \vee b = 0$$

Field

A commutative ring that each non-zero $a \in S$ has a multiplicative inverse.

Example 4.1. A field is an integral domain because if $ab = 0$ and a is non-zero then

$$a^{-1}ab = b = 0$$

4.2 Introduction

Let K be a commutative ring. Then the set of formal power series over K is

$$K[[X]] = \{a\}_i \in Ka_0 + a_1x + a_2x^2 + \dots$$

The addition and multiplication can be generalized from $K[X]$, the set of polynomials in K.

Definition (Order): Let $S \in K[[X]]$ then $\omega(S)$ is the least $n \geq 0$ such that $a_n \neq 0$. Furthermore, $\omega(0) = \infty$.

Definition (Summability): Let I be a set of indices then the family $\{S_i\}_{i \in I}$ is **summable** if for each $k \geq 0$, $w(S_i) \geq k$ for all S_i except only a finite number of them. Then we can define S to be the sum of this family

$$S = \sum_{i \in I} S_i = \sum_{i \in I} \sum_j a_j^{(i)} x^j$$

which is sensible since only a finite number of $a_j^{(i)}$ are non-zero.

Proposition 4.1. *Let K be an integral domain, then for any two formal series*

$$\omega(ST) = \omega(S) + \omega(T)$$

Asume that $\infty + k = \infty$ for any finite number k and $\infty + \infty = \infty$.

Proof. If either of S or T is zero then the proposition is true. Otherwise, both $n = \omega(S)$ and $m = \omega(T)$ are finite. Equivalently $a_n \neq 0$ and $b_m \neq 0$ while $a_k = 0, b_l = 0$ for $k < n, l < m$. Therefore,

$$ST = \sum_i \sum_{j=0}^{i} a_j b_{i-j} x^i = \sum_i c_i x^i$$

for $i < m + n$, $c_i = 0$ and $c_{m+n} = a_n b_m$ which is non-zero as K is an integral domain which was what was wanted. ■

Corollary 4.2. *For an integral domain K, the ring $K[[X]]$ is an integral domain.*

For simplicity from now on assume K is either an integral domain or a field.

Consider two formal series

$$S = \sum_n a_n x^n, \qquad T = \sum_m b_m x^m$$

with $b_0 = 0$. Then we can define the composition $S \circ T$ as follow

$$S \circ T = \sum_n a_n T^n$$

To show that it is well defined we need to show that only a finite number of T^n have degree less than k for all $k \geq 0$. Which is clear implied by the fact that $\omega(T) \geq 1$ and by Propostion 4.1, $\omega(T^n) \geq n$. It can easily be shown that

$$\begin{cases} (S_1 + S_2) \circ T & = S_1 \circ T + S_2 \circ T \\ (S_1 \cdot S_2) \circ T & = (S_1 \circ T)(S_2 \circ T) \end{cases}$$

Moreover, if $\{S_i\}$ is a summable family of formal series then

$$(\sum_i S_i) \circ T = \sum_i (S_i \circ T)$$

Proposition 4.3. *For three formal series $S, T,$ and U with $\omega(T), \omega(U) \geq 1$*

$$(S \circ T) \circ U = S \circ (T \circ U)$$

Proposition 4.4. *A formal series $S = \sum_i a_i x^i$ has a multiplicative if and only if $a_0 \neq 0$.*

Proposition 4.5. *We have that if $\omega(T) \geq 1$*

$$\omega(S \circ T) = \omega(S)\omega(T)$$

For two formal series f, g we write $f \equiv g \pmod{T^N}$ if $a_n = b_n$ for all $n = 0, 1, \ldots, N-1$. Cleary $f \equiv g \pmod{T^N}$ for all N is equivalent to $f = g$.

Proposition 4.6. *Suppose $f_1 \equiv f_2, g_1 \equiv g_2, h_1 \equiv h_2 \pmod{T^N}$. Then*

$$f_1 + g_1 \equiv f_2 + g_2 \pmod{T^N}, \quad f_1 g_1 \equiv f_2 g_2 \pmod{T^N}, \quad f_1 \circ h_1 \equiv f_2 \circ h_2 \pmod{T^N}$$

4.3 Formal Derivative

The derivative of a formal series is defined as

$$\frac{\mathrm{d}S}{\mathrm{d}x} = a_1 + 2a_2 x + 3a_3 x + \ldots$$

Proposition 4.7. *The derivative has the following properties*

1. $(S_1 + S_2)' = S_1' + S_2'$.
2. *The Leibniz rule holds*
$$(S_1 \cdot S_2)' = S_1' S_2 + S_1 S_2'$$
3. *if S has multiplicative inverse then*
$$\left(\frac{1}{S}\right)' = -\frac{S'}{S^2}$$

4. *The chain rule holds, if* $\omega(T) \geq 1$ *then*

$$(S \circ T)' = (T' \circ S) \circ S'$$

5. *The higher order derivatives can be defined recursively*

$$S^{(n)} = (S^{(n-1)})', \qquad S^{(0)} = S$$

Then, we can write the formal Taylor series of S *as*

$$S = \sum_n \frac{S^{(n)}(0)}{n!} x^n$$

4.4 Compositional inverse

Given a formal series S we wish to find a formal series T such that

1. $\omega(T) \geq 1$.
2. $S \circ T = X$.

Theorem 4.8. *S has compositional inverse T, if and only if $\omega(S) = 1$. In this case, T is unique and $T \circ S = X$ as well.*

4.5 Power series

A power series is a complex function of the form

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Theorem 4.9. *For every power series there exists a number $0 \leq R \leq \infty$, called the radius of convergence, with the following properties*

- *The series converges absolutely for every z with $|z| < R$ and it is an analytic function. The derivative can be obtained by termwise differentiation and it has the same radius of convergence. The convergence is uniform in $|z| \leq \rho$ for $0 \geq \rho \geq R$.*
- *If $|z| > R$ the series diverges.*

Proposition 4.10. *Let f and g be two power series that converge absolutely in $B_r(0)$. Then $f + g$, gf, and αf , $\alpha \in \mathbb{C}$ are absolutely convergent in $B_r(0)$.*

Example 4.2. Let $\alpha \in \mathbb{C}$

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}, \quad \binom{\alpha}{0} = 1$$

then the binomial formal series is defined as

$$(1+T)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} T^n$$

For non-negative integer α the radius of convergence is infinite but for other values of α the radius is 1.

Theorem 4.11 (Abel's limit theorem). *If $\sum_{n=0}^{\infty}$ converges, then $f(z) = \sum_{n=0}^{\infty} a_n z^n$ tends to $f(1)$ as z approaches 1 in such a way that $\frac{|1-z|}{1-|z|}$ remains bounded.*

Theorem 4.12.

1. *Let f be a non-constant power series with non-zero radius of convergence. If $f(0) = 0$ then there exists $s > 0$ such that $f(z) \neq 0$ for all non-zero z with $|z| \leq s$.*
2. *Let f and g to be two convergent power series. Suppse $f(z) = f(z)$ for all points x in an infinite set having 0 as a point of accumulation. Then $f = g$.*

Let $f = \sum a_n z^n$ and $\phi = \sum c_n z^n$ with $c_n \in \mathbb{R}_+$ be two formal power series. Then f is dominated by ϕ

$$f \prec \phi$$

if $|a_n| \leq c_n$ for all n.

Proposition 4.13. *Suppose $f \prec \phi$ and $g \prec \psi$ then*

$$f + g \prec \phi + \psi, \quad fg \prec \phi\psi$$

Theorem 4.14. *Let f be a power series with non-zero constant term and non-zero radius of convergence and let g be the inverse of f that is, $fg = 1$. Then g has a non-zero radius of convergence.*

Proposition 4.15. *Let*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad h(z) = \sum_{n=1}^{\infty} b_n z^n$$

be two power series and f is absolutely convergent for $|z| \leq r$ for some $r > 0$ and let $s > 0$ be such that

$$\sum_{n=1}^{\infty} |b_n| s^n \leq r$$

Then $f \circ h$ converges absolutely for $|z| \leq s$.

4.6 Analytic functions

Let f be defined in a neighbourhood of z_0. f is said to be analytic if there exists some power series

$$\sum a_n (z - z_0)^n$$

which converges absolutely for $|z - z_0| < r$ and for such z

$$f(z) = \sum a_n (z - z_0)^n$$

Proposition 4.16. *If f and g are analytic on U then $f + g$, fg, f/g for $g(z) \neq 0$ are also analytic. If $g : U \to V$ and $f : V \to \mathbb{C}$ then $f \circ g$ is analytic.*

Theorem 4.17. *Let $f(z) = \sum a_n z^n$ with radius of convergence of r. Then f is analytic on $B_r(0)$.*

Theorem 4.18. *Let $f(z) = \sum a_n z^n$ with radius of convergence of r then*

1. *The series $\sum n a_n z^{n-1}$ has the same radius of convergence.*
2. *f is holomorphic on $B_r(0)$ and its derivative is equal to $\sum n a_n z^{n-1}$.*

We shall see later that every holomorphic function admits a power series.

4.7 Inverse and open mapping

f is analytic isomorphism if $f : U \to V$ is analytic and V is open and there exists analytic function $g : V \to U$ such that $f \circ g = \mathrm{id}_V$ and $g \circ f = \mathrm{id}_U$. f is locally analytic isomorphism at z_0 or locally invertible if there exists an open neighbourhood U containing z_0 such that f is an analytic isomorphism on U.

Theorem 4.19. *Let $f(z) = a_1 z + \dots$ be a formal power series with $a_1 \neq 0$. We know there exists a unique $g(z)$ such that $f(g(z)) = z$ and $g(f(z)) = z$. If f is convergent a power series, then g is a convergent power series as well.*

Theorem 4.20. *Let f be an analytic function on open set U containing z_0. Suppose that $f'(z_0) \neq 0$ then, f is a local analytic isomorphism at z_0.*

Definition: Let U be an open set then $f : U \to V$ is open mapping if for each open subset $U' \subset U$, $f(U')$ is open.

Theorem 4.21. *Let f be an analytic function on open set U such that for each point of U, f is not constant on a given neighbourhood of that point. Then f is an open mapping.*

Theorem 4.22. *Let f be an analytic function on open set U and f is injective. Let $V = f(U)$ then, $f : U \to V$ is an analytic isomorphism and $f'(z) \neq 0$ for all $z \in U$.*

4.8 The local maximum modulus principle

Definition: f is locally constant at a point z_0 if there exists an open set D containing z_0 such that f is constant on D.

Theorem 4.23. *Let f be an analytic function on open set U. Let $z_0 \in U$ be a maximum for $|f|$, that is*

$$|f(z_0)| \geq |f(z)| \quad \forall z \in U$$

Then f is locally constant at z_0.

Corollary 4.24. *Above theorem also holds for $\Re$ instead of $|\cdot|$.*

Theorem 4.25. *Let f be a non-constant polynomial*

$$f(z) = a_0 + a_1 z + \dots + a_d z^d$$

with $a_d \neq 0$. Then f has some complex roots.

Chapter 5

Complex Integrals

5.1 Integrals over paths

Path is a sequence of regular curves

$$\gamma = \{\gamma_1, \ldots, \gamma_n\}$$

such that

$$\gamma_j(b_j) = \gamma_{j+1}(a_{j+1})$$

Theorem 5.1. *Suppose U is a connected open set and f is holomorphic on U. If $f' = 0$ then f is constant on U.*

Definition: If f is a function on an open set Ω and g is a holomorphic function on Ω, such that $g' = f$, then g is called a *primitive* if f on Ω.

Corollary 5.2. *On an connected open set U, the primitives of are determined up to a constant.*

Theorem 5.3. *Suppose U is a connected open set then*

1. *If f is analytic on U and not constant, the set of zeroes of f on U is discrete.*
2. *f, g are analytic on U and $S = \{z \mid f(z) = g(z)\}$ is not discrete then $f = g$ on U.*

Let $f : [a, b] \to \mathbb{C}$ be a continuous function with

$$f(t) = u(t) + iv(t)$$

then

$$\int f(t)\,\mathrm{d}t = \int u(t)\,\mathrm{d}t + i\int v(t)\,\mathrm{d}t$$

and by the Fundamental theorem of calculus

$$G(t) = \int_a^t F(\tau)\,\mathrm{d}\tau$$

is differentiable with $G'(t) = F(t)$. We can then expand the notion of complex integrability to continuous complex function $f : U \to \mathbb{C}$, where U is an open subset of $\mathbb{C}$ using regular curves such as $\gamma : [a, b] \to U$. The integral of f along γ is

$$\int_\gamma f(t) = \int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t$$

This definition of integral is invariant to substitution. To see this, let $g : [a, b] \to [c, d]$ be a $\mathcal{C}^1$ function and $\gamma = \psi \circ g$

$$\begin{aligned}\int_\gamma f &= \int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t \\ &= \int_a^b f(\psi(g(t)))\psi'(g(t))g'(t)\,\mathrm{d}t \\ &= \int_c^d f(\psi(\tau))\psi'(\tau)\,\mathrm{d}\tau \\ &= \int_\psi f\end{aligned}$$

We can further expand this notion to paths $\gamma = \{\gamma_1, \ldots, \gamma_n\}$ as follow

$$\int_\gamma f = \sum_{i=1}^n \int_{\gamma_i} f$$

Theorem 5.4. *Let f be a continuous function on open set U with primitive g. Let $\gamma : [a, b] \to U$ be a path with $\alpha = \gamma(a)$ and $\beta = \gamma(b)$ then*

$$\int_\gamma f = g(\beta) - g(\alpha)$$

Corollary 5.5. *If γ is a closed path in U then*

$$\int_\gamma f = 0$$

Theorem 5.6. *Let U be a connected open set and f a continuous function U. If for any closed path γ on U, the integral of f along γ is zero , $\int_\gamma f = 0$, then f has a primitive g on U.*

Proof. Pick $z_0 \in U$ and define

$$g(z) = \int_{z_0}^z f$$

where the integral is taken over any path from z_0 to z. Let γ, η be two curves from z_0 to z and let η^- be the reverse of η. Since

$$\int_\gamma f + \int_{\eta^-} f = 0 \implies \int_\gamma f = \int_\eta f$$

and hence g is well-defined. Consider the difference quotient

$$\frac{g(z+h) - g(z)}{h} = \frac{1}{h}\int_z^{z+h} f$$

where the integral can be taken along any segments. Furthermore, we can write

$$f(\xi) = f(z) + \phi(\xi)$$

where $\lim_{\xi \to z} \phi(\xi) = 0$. Then,

$$\begin{aligned}
\frac{1}{h}\int_z^{z+h} f(\xi)\,\mathrm{d}\xi &= \frac{1}{h}\int_z^{z+h} f(z)\,\mathrm{d}\xi + \frac{1}{h}\int_z^{z+h} \phi(\xi)\,\mathrm{d}\xi \\
&= f(z)\frac{1}{h}\int_z^{z+h} \phi(\xi)\,\mathrm{d}\xi \\
&\leq f(z) + \max_{\xi}|\phi(x)| \xrightarrow{h\to 0} f(z) \\
\implies g' &= f
\end{aligned}$$

Let $\gamma : [a, b] \to \mathbb{C}$ be a regular curve. The *speed* is of γ is defined as $|\gamma'|^2$ and the length of γ is

$$l(\gamma) = \int_a^b |\gamma'(t)|\,\mathrm{d}t$$

and for a path $\gamma = \{\gamma_1, \ldots, \gamma_n\}$

$$l(\gamma) = \sum_{i=1}^{n} l(\gamma_i)$$

and sup norm of f on a set S

$$\|f\| = \sup_{z\in S}|f|$$

and over a curve γ the sup-norm of f is

$$\|f\|_\gamma = \sup_{t\in[a,b]} |f(\gamma(t))|$$

Proposition 5.7. *ML inequality Let f be a continuous function on U and γ be a path in U them*

$$\left|\int_\gamma f\right| \leq \|f\|_\gamma l(\gamma)$$

Theorem 5.8. *Let $\{f_n\}$ be a sequence of continuous functions on U with $f_n \rightrightarrows f$ then*

$$\lim_{n\to\infty}\int_\gamma f_n = \int_\gamma f$$

and if $\sum f_n$ is a series of continuous function which converges uniformly on U

$$\int_\gamma \sum f_n = \sum \int_\gamma f_n$$

5.2 Local primitive for a holomorphic function

Theorem 5.9 (Goursat). *Let R be a rectangle, and let f be a holomorphic function on R then*

$$\int_{\partial R} f = 0$$

Theorem 5.10. *Let U be a disc centered at a point z_0 and f a continuous function on U. Assume that for any rectangle $R \subset U$*

$$\int_{\partial R} f = 0$$

For each point z in disk define

$$g(z) = \int_{z_0}^{z} f$$

where the integral is taken over a rectangle R whose opposite vertices are z_0 and z. Then, g is holomorphic on U and is a primitive for f.

Theorem 5.11. *Let U be a disc and f be a holomorphic function on U. Then, f has a primitive on U and the integral of f along any closed path in U is zero.*

5.3 Path integrals for continuous curves

Knowing that local primitive exists for holomorphic functions allows us to describe their integral along a path in a way which makes no use of differentiability of the path and applys to continuous paths.z

Lemma 5.12. *Let $\gamma : [a, b] \to U$ be a continuous curve in an open set U. Then, there exists some positive number $r > 0$ such that every point on the curve lies at distance $\geq r$ from the complement of U.*

Proof. Consider

$$\phi(t) = \min_{w \in U^c} |\gamma(t) - w|$$

$|\gamma(t) - w|$ is continuous in terms of w by considering a sufficiently large disc that contains U we can bound the domain of w to the intersection of this disc and U^c. Thus, due to compactness of the intersection, $|\gamma(t) - w|$ achieves its minimum. Moreover, since $\phi(t)$ is continuous and its domain is compact then, it has a minimum and the minimum is non-zero since U is open. ■

Definition: Let $P = \{a_0, \ldots, a_n\}$ be a partition of $[a, b]$ and $\{D_0, \ldots, D_n\}$ be a sequence of discs. This sequence is *connected* by curve along the partition P if D_i contains $\gamma([a_i, a_{i+1}])$.

Let $\epsilon < \frac{r}{2}$ where r is the same as preceding lemma. Since γ is uniformly continuous, there exists a δ such that

$$\forall t, s \in [0, 1],\ |t - s| < \delta \implies |\gamma(t) - \gamma(s)| < \epsilon$$

Consider a partition P on $[a, b]$ such that $\|P\| < \delta$. Then, the $\gamma([a_i, a_{i+1}])$ lies in the disc $D_i = B_\epsilon(a_i)$. Let f be a holomorphic function on U.

$$\int_\gamma f = \sum_{i=0}^{n-1} \int_{\gamma_i} f \qquad \gamma_i = \gamma([a_i, a_{i+1}])$$

Let $z_i = \gamma(a_i)$ and g_i be a primitive of f on disc D_i. If each $\gamma_i \in \mathcal{C}^1$ then we know that

$$\int_\gamma f = \sum_{i=0}^{n-1} g_i(z_{i+1}) - g_i(z_i)$$

By considering the following lemma, we can relax the regularity condition for γ and reduce it to continuity.

Lemma 5.13. *Let U be an open set and $\gamma : [a, b] \to U$ be a continuous curver. Let*

$$a = a_0 \leq a_1 \leq \cdots \leq a_n = b$$

be a partition on $[a, b]$ such that $\gamma([a_i, a_{i+1}])$ is contained in a disc D_i which itself is contained in U. Let f be a holomorphic function on U, g_i be a primitive of f on D_i, and $z_i = \gamma(a_i)$, then

$$\sum_{i=0}^{n-1} g_i(z_{i+1}) - g_i(z_i)$$

is independent of the choices of partition, discs D_i, and primitives g_i on D_i subjected to stated condition. Therefore, Lemma 5.13 depends only on γ and hence the integral on continuous path is well-defined.

Proof.

Definition: Let γ, η be two paths defined on $[a, b]$. We say that they are *close* together if there exists a partition

$$a = a_0 \leq a_1 \leq \cdots \leq a_n = b$$

and for each $i = 0, \ldots, n-1$ there exists a disc D_i contained in U such that

$$\gamma([a_i, a_{i+1}]), \eta([a_i, a_{i+1}]) \subset D_i$$

Lemma 5.14. *Let γ, η be continuous paths on open set U, that are close together and have the same endpoints. Let f be holomorphic on U*

$$\int_\gamma f = \int_\eta f$$

Proof.

5.4 Homotopy

Let $\gamma, \eta : [a, b] \to U$ be two paths. γ is *homotopic* to η, if there eists a continuous function

$$\Psi : [a, b] \times [c, d] \to U$$

such that

$$\Psi(t, c) = \gamma(t) \qquad \Psi(t, c) = \eta(t)$$

for all $t \in [a, b]$. Intuitively, Ψ is a continuous deformation of γ to η. Ψ *leave the end point fixed* if we have

$$\Psi(a, s) = \gamma(a) \qquad \Psi(b, s) = \gamma(b)$$

for all $s \in [c, d]$. Similarly, we assume that a homotopy of two closed paths is such that each path $\Psi(\,\cdot\,, s)$ is a closed path.

Theorem 5.15. *Let γ, η be two homotopic continuous paths on open set U with the same endpoints and f be a holomorphic function on U.*

$$\int_\gamma f = \int_\eta f$$

In particular, if γ, η are closed paths and homotopic to a point in U then

$$\int_\gamma f = \int_\eta f = 0$$

Proof. Let $\Psi : [a, b] \times [c, d] \to U$ be the homotopy of the closed paths γ, η. Since Ψ is a continuous function on a compact domain, the image of Ψ is compact and hence it has a positive distance r from U^c. Similarly, by considering the uniform continuity we can have partitions $P = \{a_1, \ldots, a_n\}$ on $[a, b]$ and $Q = \{c_1, \ldots, c_m\}$ on $[c, d]$ and

$$S_{ij} = [a_i, a_{i+1}] \times [c_j, c_{j+1}]$$

such that $\Psi(S_{ij})$ is contained in D_{ij} which itself is contained in U. Let Ψ_j

$$\Psi_j(t) = \Psi(t, c_j)$$

Then Ψ_j and Ψ_{j+1} are closed together and by applying the preceding lemma we get

$$\int_\gamma f = \int_{\Psi_0} f = \int_{\Psi_1} f = \cdots = \int_{\Psi_m} f = \int_\eta f$$

A set S of complex numbers is conex, if for any $z, w \in S$ the segment $[z, w, \subset]S$. For examples, discs and rectangles are convex.

Lemma 5.16. *Let S be a convex set and γ, η continuous closed curves in S. Then, γ, η are homotopic in S.*

Proof.

$$\Psi(t, s) = s\gamma(t) + (1 - s)\eta(t)$$

Open set U is simply connected if it is connected and every closed path is homotopic to a point in U. By the preceding lemma, every convec set is simply connected, as path connectednes implies connectedness.

5.5 Existence of global primitives

Theorem 5.17. *Let f be a holomorphic function on a simply connected open set U and let $z_0 \in U$. For an $z \in U$*

$$g(z) = \int_z^{z_0} f(\xi)\,\mathrm{d}\xi$$

is independent of the path in U and g is a primitive for f.

Example 5.1. Let U be a simply connected open set not containing 0. Pick $z_0 \in U$ and w_0 such that $e^{w_0} = z_0$, define

$$\log z = w_0 + \int_{z_0}^{z} \frac{1}{\xi}\,\mathrm{d}\xi$$

Then, log is a primitive for $\frac{1}{z}$. If $L(z)$ is another primitive for $\frac{1}{z}$ on U such that $e^{L(z)} = z$, then there exists an integer k such that

$$L(z) = \log z + 2\pi i k$$

5.6 Local Cauchy formula

Theorem 5.18 (Local Cauchy formula). *Let $\bar{D}$ be a closed disc of positive radius and f holomorphic on $\bar{D}$ (open disc U containing $\bar{D}$). Let γ be the circle with is the boundary of $\bar{D}$. Then, for every $z_0 \in D$*

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(\xi)}{\xi - z_0} \, \mathrm{d}\xi$$

Theorem 5.19. *Let f be holomorphic on open set U then f is analytic on U.*

Theorem 5.20. *Let f be holomorphic on closed disc $\bar{B}_R(z_0)$, $R > 0$. Let C_R be the circle bounding the disc. Then, f has a power series expansion*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

whose coefficients are given by the formula

$$a_n = \frac{1}{n!} f^{(n)}(z_0) = \frac{1}{2\pi i} \int_{C_R} \frac{f(\xi)}{(\xi - z_0)^{n+1}} \, \mathrm{d}\xi$$

Furthermore, if $\|f\|_{C_R}$ denotes the sup-norm of f on C_R then

$$|a_n| = \frac{\|f\|_{C_R}}{R^n}$$

in particular, the radus of convergence of the series is greater than R.

The preceding theorems imply that a function is analytic if and only if it is holomorphic.

Definition: A function f is entire if it is holomorphic on all $\mathbb{C}$. Hence, entire functions have convergence radius of ∞.

Corollary 5.21. *Let f be entire and $\|f\|_R$ be its sup-norm on the circle of radius R. Suppose that there exists a constant C and a positive integer k such that*

$$\|f\|_R = CR^k$$

for arbitrarily large R. Then, f is a polynomial of degree $\leq k$.

Theorem 5.22 (Liouville's theorem). *A bounded entire function os constant.*

Corollary 5.23. *A polynomial over complex number which does not have root in $\mathbb{C}$ is constant.*

Theorem 5.24. *Let γ be a path in an oopen set U and g be a continuous function on γ. If z is not on γ define*

$$f(z) = \int_\gamma \frac{g(\xi)}{\xi - z} \, \mathrm{d}\xi$$

Then f is analytic on the complement of γ in U and its derivatives are given by

$$f^{(n)}(z) = n! \int_\gamma \frac{g(\xi)}{(\xi - z)^{n+1}} \, \mathrm{d}\xi$$

Corollary 5.25. *Let f be an analytic function on closed disc $\bar{B}_R(z_0)$ with $R > 0$. Let $0 < R_1 < R$ and denote the sup-norm of f on the circle of radius R by $\|f\|_R$. For $z \in \bar{B}_R(z_0)$*

$$\left|f^{(n)}(z)\right| \leq \frac{n!R}{(R-R_1)^{n+1}}\|f\|_R$$

Theorem 5.26 (Morera's theorem). *Let U be an open set in $\mathbb{C}$ and let be continuous on U. Assume that the integral of f along the boundary of every rectangle in U is zero. Then, f is analytic.*

Chapter 6

Winding Numbers

6.1 Winding number

Definition: Winding number of a closed path γ with respect to a point α is

$$W(\gamma, \alpha) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z - \alpha} \, \mathrm{d}z = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(t)}{\gamma(t) - \alpha} \, \mathrm{d}t$$

provided that γ does not pass through α.

Lemma 6.1. *If γ is a closed path, then $W(\gamma, \alpha)$ is an integer.*

Proof. Consider a the integral and compare it to $\gamma(t) - \alpha$. ■

Lemma 6.2. *Let γ be a path. Then, the function of α defined by*

$$\alpha \mapsto \int_\gamma \frac{1}{z - \alpha} \, \mathrm{d}z$$

for α no on the path, is a continuous function of α.

Lemma 6.3. *Let γ be a closed path and S a connected set not intersecting γ. Then,*

$$\alpha \mapsto \int_\gamma \frac{1}{z - \alpha} \, \mathrm{d}z$$

is a constant for α in S. If S is not bounded, then this constant is zero.

Definition: A closed path γ in U is homologous to 0 in U if

$$\int_\gamma \frac{1}{z - \alpha} \, \mathrm{d} = 0$$

for every point α not in U. Equivalently,

$$W(\gamma, \alpha) = 0 \quad \forall \alpha \in U^c$$

Furthermore, two closed paths γ, η are homologous in U if

$$W(\gamma, \alpha) = W(\eta, \alpha) \quad \forall \alpha \in U^c$$

Theorem 6.4. *1. If γ, η are closed paths in U and homotopic then they are homologous. Converse is true if their inside is contained in U.*

2. If γ, η are closed paths and close together in U then they are homologous.

Definition: Let $\gamma_1, \ldots, \gamma_n$ be a sequence of curves and let $m_1, \ldots, m_n$ be integers. The formal sum

$$\gamma = m_1\gamma_1 + \cdots + m_n\gamma_n = \sum_{i=1}^{n} m_i\gamma_i$$

is called a chain. We define

$$\int_\gamma f = \sum_{i=1}^{n} m_i \int_{\gamma_i} f$$

and thus

$$W(\gamma, \alpha) = \frac{1}{2\pi i}\int_\gamma \frac{1}{z-\alpha}\,\mathrm{d}z = \sum_{i=1}^{n} m_i W(\gamma_i, \alpha)$$

Proposition 6.5. *If γ, η are closed chains in U then*

$$W(\gamma + \eta, \alpha) = W(\gamma, \alpha) + W(\eta, \alpha)$$

γ and η are homologous in U if

$$funcW\gamma, \alpha = W(\eta, \alpha) \quad \forall \alpha \in U^c$$

6.2 Cauchy's formula

Theorem 6.6 (Cauchy's theorem). *Let γ be a closed chain an open set U such that γ is homologous to zero in U. Let f be holomorphic in U. Then*

$$\int_\gamma f = 0$$

Corollary 6.7. *If γ, η are closed chains in U then*

$$\int_\gamma f = \int_\eta f$$

Theorem 6.8. *Let U be an open set and γ be a closed chain in U homologous to zero. Let $z_1, \ldots, z_n$ be finite number of distinct points of U. Let γ_i be the boundary of closed disc $\bar{D}_i$ contained in U, containing z_i, and oriented counter-clockwise. We assume that $\bar{D}_i$ does not intersect $\bar{D}_j$ if $i \neq j$. Let*

$$m_i = W(\gamma, z_i)$$

Let U^* be the set U without $\{z_1, \ldots, z_n\}$. Then γ is homologous to $\sum m_i\gamma_i$ in U^*.

Theorem 6.9 (Cauchy's formula). *Let γ be a closed chain in open set U homologous to zero in U. Let f be an analytic function on U and $z_0 \in U$ not on γ*

$$\frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z - z_0}\,\mathrm{d}z = W(\gamma, z_0)f(z_0)$$

Proof. use analytic ■

Proof. Dixon, define $g : U \times U$

$$(w, z) \mapsto \begin{cases} \dfrac{f(w) - f(z)}{w - z} & w \neq z \\ f'(z) & w = z \end{cases}$$

for each w, $z \mapsto g(w, z)$ is analytic. g is continuous, define a bounded entire function. ■

6.2.1 Artin's proof

Lemma 6.10. *Let γ be a path in open U. Then, there exists a rectangular path η with the same endpoints such that γ and η are close together. In particular, γ and η are homologous in U and for any holomorphic function f on U we have*

$$\int_\gamma f = \int_\eta f$$

This lemma reduces Cauchy's formula to the case when γ is rectangular.

Definition: Let $\gamma_i = \gamma([a_i, a_{i+1}])$ then the chain

$$\gamma_1 + \cdots + \gamma_n$$

is a subdivision of γ. If η_i are obtained by a reparameterization of γ_i the

$$\eta_1 + \cdots + \eta_n$$

are also subdivision of γ.

Theorem 6.11. *Let γ be a rectangular closed chain in U and homologous to zero. Then there exists rectangles $R_1, \ldots, R_N$ contained in U such that if ∂R_i is the boundary of R_i oriented in counter-clockwise then*

$$\sum_{i=1}^{N} m_i \partial R_i$$

for some integers m_i is a subdivision of γ.

Proof. $m_i = W(\gamma, \alpha_i)$, $\alpha_i \in R_i$. If $m_i \neq 0$ for R_i then $R_i \subset U$. $\sum m_i \partial R_i$ is a subdivision. ■

Chapter 7

Application of Cauchy's formula

7.1 Uniform limits of analytical functions

Theorem 7.1. *Let $\{f_n\}$ be a sequence of holomorphic functions on a open set U. Assume that for each compact subset K of U, the sequence converges uniformly on K and let the limit function be f. Then f is holomorphic.*

Theorem 7.2. *Let $\{f_n\}$ be a sequence of holomorphic functions on a open set U. Assume that for each compact subset K of U, the sequence converges uniformly on K and let the limit function be f. Then the sequence of derivatives converges uniformly on every compact subset K, $f' = \lim f_n'$.*

7.2 Laurant seris

The series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

is the Laurant series expansion. Let A be a set of complex number. Laurant series converges absolutely (uniformly) on A if the two series

$$f^+(z) = \sum_{n\geq 0} a_n z^n \qquad f^-(z) = \sum_{n<0} a_n z^n$$

converge absolutely (uniformly). In that case

$$f = f^- + f^-$$

Theorem 7.3. *Let A be the annulus $A = \{z \,|\, r \leq |z| \leq R\}$ for some $r \leq R$. Let f be holomorphic on A and $r < s < S < R$. Then f has Laurant expansion*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

where

$$a_n = \begin{cases} \frac{1}{2\pi i}\int_{C_R} \frac{f(\xi)}{\xi^{n+1}} & n \geq 0 \\ \frac{1}{2\pi i}\int_{C_r} \frac{f(\xi)}{\xi^{n+1}} & n < 0 \end{cases}$$

converges absolutely on $s \leq |z| \leq S$.

7.3 Singularity

Let D be an open disc cenetered at z_0 and $U = D/\{z_0\}$. Let f be analytic on U. Then, f is said to have isolated singularity at z_0.

7.3.1 Removable singularity

Theorem 7.4. *If f is bounded in some neighbourhood of z_0, then one can define $f(z_0)$ in a uninque way such that f is analytic on z_0*

Proof. Laurant expansion ■

This kind of singularity is called Removable singularity.

7.3.2 Poles

If f has finite negative terms in its Laurant expansion

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + a_0 + a_1(z - z_0) + \dots$$

Then f is said to have a pole of order m. However, the order of f at z_0 is $-m$.

$$\operatorname{ord}_{z_0} fg = \operatorname{ord}_{z_0} f + \operatorname{ord}_{z_0} g$$

Proposition 7.5. *f has a pole of order m at z_0 if and only if $f(z)(z - z_0)^m$ is holomorphic at z_0 and has no zero at z_0.*

Definition: f i defined on U/S, where S is a discrete set of points which are the poles of f, is mermorphic on U. Thus, it is the quotient of two holomorphic functions in the neighbourhood of a point.

7.3.3 Essential singularity

When the Laurant expansion has inifinite negative terms.

Theorem 7.6 (Casorati-Weirestrass). *Let 0 be an essential singularity of the function f. Let D be a disc cenetered at 0 on which f is holomorphic except at 0. Let U be the complement of $\{0\}$ in D. Then, $f(U)$ is dense in the complex numbers. In other words, the values of f on U come arbitrarily close to any complex number. In fact f takes on every complex value except possible one.*

Theorem 7.7. *The only analytic automorphism of $\mathbb{C}$ are functions of the form $f(z) = az+b$, where a, b are constants and $a \neq 0$.*